

Machine Learning–Based Spatio-Temporal Excess Path Loss Modeling for LEO Satellite–Ground Links

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Abstract—Low Earth Orbit (LEO) satellite communication links exhibit strong spatio-temporal variations due to rapidly changing geometry, elevation-dependent propagation conditions, and atmospheric effects. Classical analytical propagation models, while physically grounded, often struggle to capture such dynamic variations accurately. This paper presents a physics-aware machine learning framework for modeling excess path loss in LEO satellite–ground links at S-band frequencies. Free space path loss is computed analytically using slant range geometry, while the residual excess loss caused by atmospheric and environmental effects is learned using a supervised regression model. The proposed approach incorporates elevation angle, slant range, Doppler-related proxies, temporal pass indices, and weather parameters as input features. A temporal train–test strategy is adopted to evaluate generalization across unseen satellite passes. Experimental results demonstrate low prediction error with physically consistent behavior, particularly improved accuracy at moderate and high elevation angles. The results indicate that combining analytical propagation modeling with data-driven learning offers an effective and interpretable approach for spatio-temporal satellite channel characterization.

Index Terms—LEO satellite communication, excess path loss, RF propagation, machine learning, spatio-temporal modeling, S-band

I. INTRODUCTION

The rapid deployment of Low Earth Orbit (LEO) satellite constellations has renewed interest in accurate and dynamic satellite–ground propagation modeling. Unlike geostationary links, LEO satellite channels are characterized by rapidly varying geometry, time-dependent elevation angles, and non-stationary atmospheric effects. Traditional analytical propagation models, such as free space path loss and elevation-dependent attenuation formulations, provide physical interpretability but often fail to capture complex residual variations under realistic conditions.

Recent studies have explored machine learning techniques for radio propagation modeling; however, many approaches either ignore underlying physical principles or directly learn total path loss without separating deterministic and stochastic components. Such formulations reduce interpretability and limit generalization, particularly for satellite links with strong geometric dynamics.

This work proposes a physics-aware machine learning framework that explicitly separates analytically modeled free

space path loss from excess propagation loss. The primary contributions of this paper are as follows:

- A spatio-temporal dataset generation framework for LEO satellite–ground links incorporating geometry, atmospheric, and temporal features.
- An explicit excess path loss formulation that preserves physical interpretability.
- A temporal generalization evaluation strategy across unseen satellite passes.
- A detailed error analysis linking model behavior to elevation-dependent propagation characteristics.

II. RELATED WORK

A. Analytical Satellite Propagation Models

Classical satellite propagation models rely on free space path loss formulations combined with elevation-dependent atmospheric attenuation models. These approaches are computationally efficient and physically interpretable but are typically limited by simplified assumptions and static parameterization.

B. Machine Learning in Propagation Modeling

Machine learning has been widely applied to terrestrial propagation modeling, often achieving high accuracy. However, many studies employ random data splitting and directly predict total path loss, limiting physical interpretability and temporal robustness.

C. Research Gap

Existing literature lacks physics-aware machine learning formulations that explicitly separate deterministic and stochastic loss components while evaluating temporal generalization for satellite passes. This work addresses this gap.

III. SATELLITE GEOMETRY AND DATASET GENERATION

A semi-synthetic dataset is generated to model LEO satellite–ground link geometry. Satellite elevation angles are sampled between low and high elevation regimes, and corresponding slant ranges are computed based on Earth radius and satellite altitude assumptions. A Doppler-related proxy and a normalized time-of-pass index are included to capture temporal dynamics. Weather-related parameters such as rain rate, humidity, and temperature are incorporated to represent atmospheric variability.

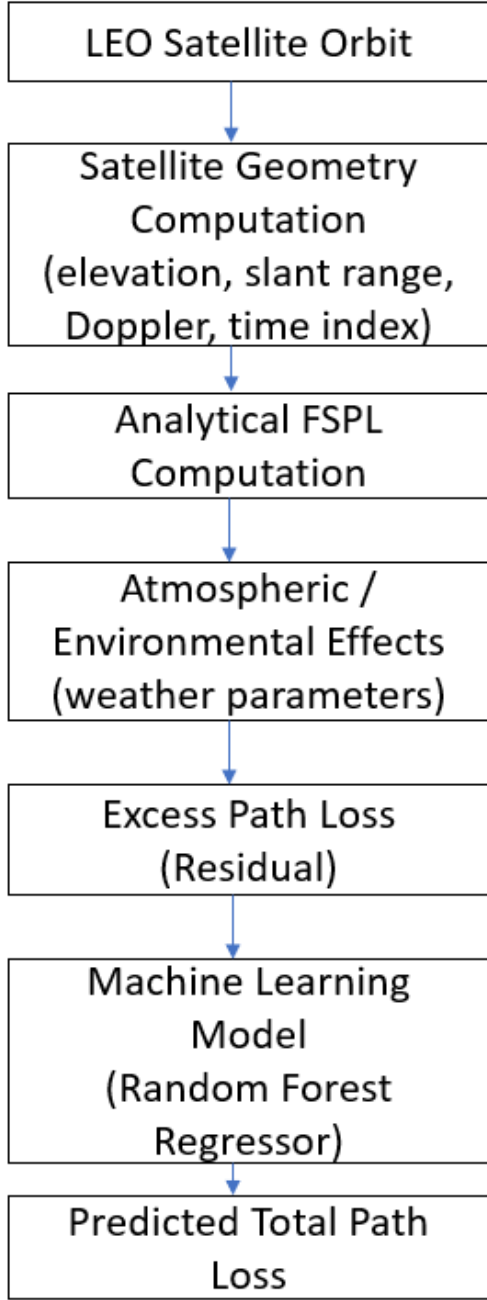


Fig. 1. Overview of the proposed spatio-temporal propagation modeling framework.

IV. PROPAGATION LOSS MODELING

A. Free Space Path Loss

Free space path loss is computed analytically using slant range geometry and operating frequency. This component captures the deterministic distance-dependent attenuation inherent to satellite links.

B. Atmospheric and Excess Loss Modeling

Elevation-dependent atmospheric loss is modeled using simplified analytical expressions that account for increased

attenuation at lower elevation angles, combined with rain-related attenuation proxies. The excess loss is defined as the difference between total modeled loss and free space path loss, isolating stochastic propagation effects.

V. MACHINE LEARNING FRAMEWORK

The machine learning model is trained to predict excess path loss using geometry, atmospheric, and temporal features. A Random Forest regressor is employed due to its robustness and interpretability. Importantly, a temporal train-test split is used, where earlier satellite passes are used for training and later passes for testing, ensuring realistic generalization evaluation.

VI. RESULTS AND EVALUATION

Model performance is evaluated using root mean square error (RMSE) and mean absolute error (MAE). The proposed approach achieves low prediction error while maintaining physically consistent behavior.

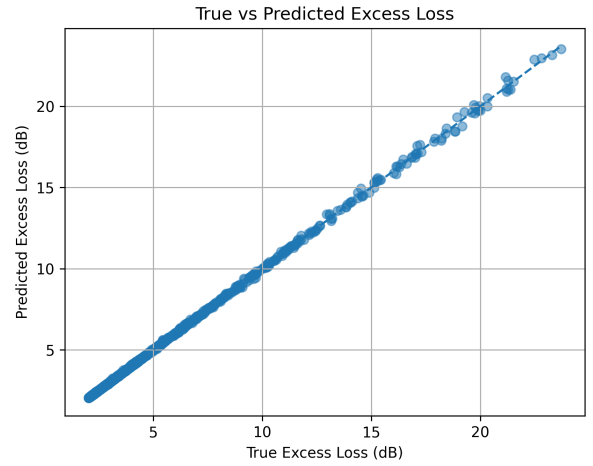


Fig. 2. True versus predicted excess path loss.

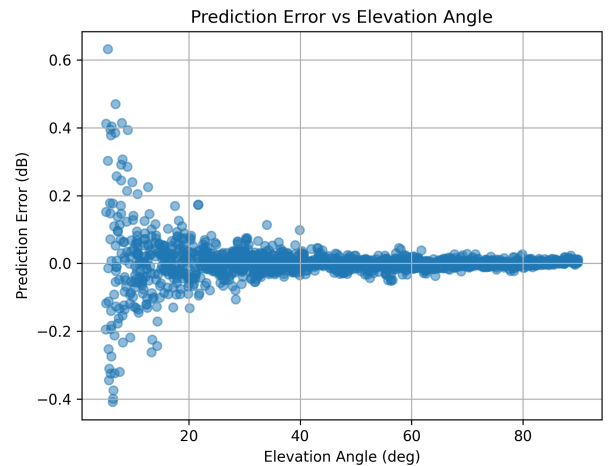


Fig. 3. Prediction error as a function of elevation angle.

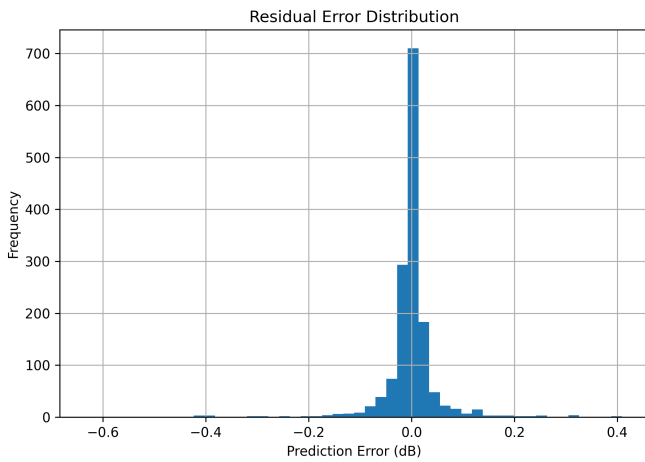


Fig. 4. Residual error distribution of excess path loss prediction.

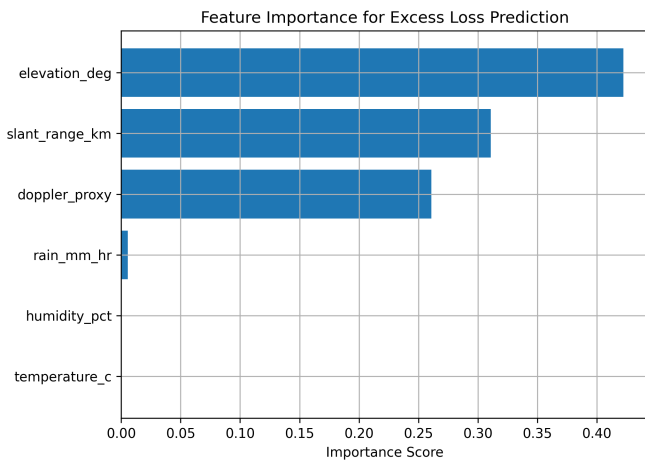


Fig. 5. Feature importance analysis of the trained model.

The results show increased error variance at low elevation angles due to longer atmospheric paths, while errors remain tightly bounded at moderate and high elevations. Feature importance analysis confirms elevation angle and slant range as dominant contributors, aligning with physical expectations.

VII. DISCUSSION

The proposed framework demonstrates that combining analytical propagation modeling with machine learning enables accurate and interpretable excess loss prediction for LEO satellite links. Temporal generalization results indicate robustness across unseen passes, while error trends reflect realistic propagation behavior. The primary limitation of this study lies in the use of semi-synthetic data, which motivates future validation using real measurement datasets.

VIII. CONCLUSION AND FUTURE WORK

This paper presented a physics-aware machine learning approach for spatio-temporal excess path loss modeling in LEO satellite-ground links. By explicitly separating free space

and excess loss components and incorporating temporal evaluation, the proposed method achieves accurate and physically consistent predictions. Future work will extend the framework to multi-band analysis and real-world measurement validation.

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